

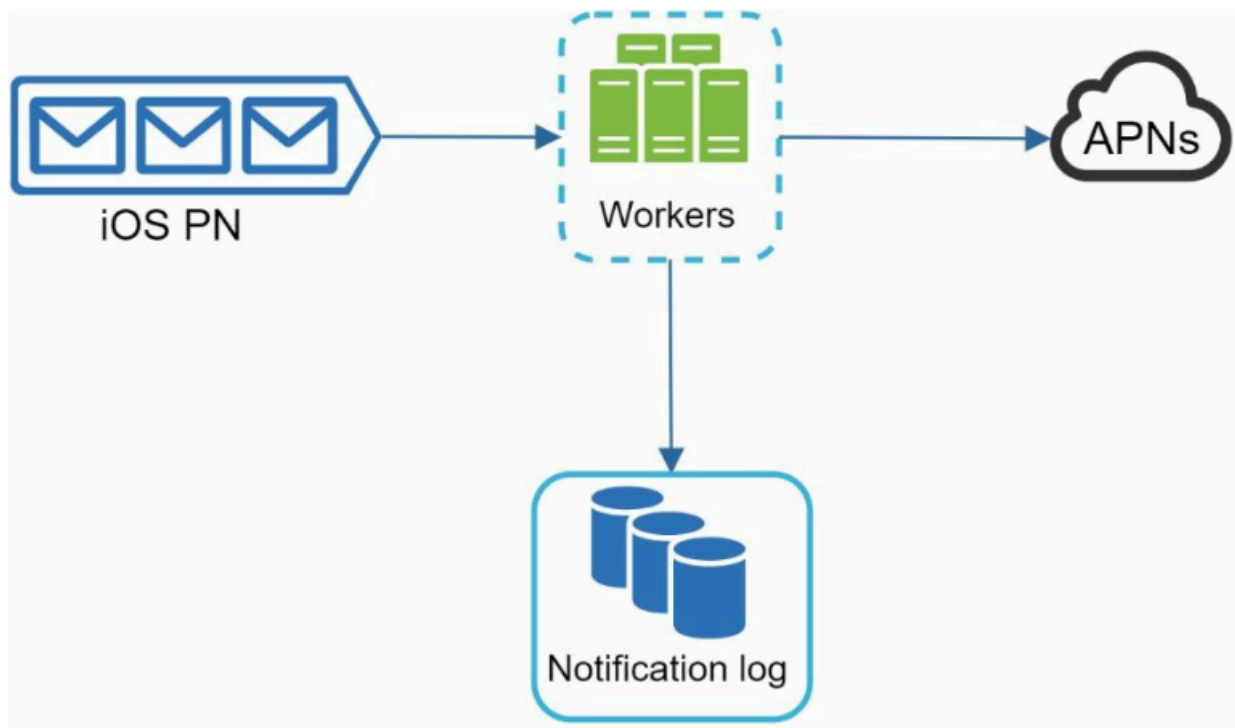
Step 3 – Design Deep Dive

In the high-level design, we discussed:

- Different types of notifications
- Contact information gathering flow
- Notification sending/receiving flow

In this section, we dive deeper into important system design considerations:

- Reliability
- Additional components and considerations
- Updated design



Reliability

Reliability is one of the most important aspects of a distributed notification system.

The system must ensure that notifications are:

- Not lost
- Delivered reliably
- Handled properly during failures

Two major reliability questions must be answered.

1. How to Prevent Data Loss?

One of the most critical requirements is:

None

Notifications can be delayed or reordered, but they should never be lost.

In distributed systems:

- Servers can crash
- Queues can fail
- Third-party services can become unavailable
- Network failures can happen

Thus, the system must guarantee data persistence.

Solution: Persist Notification Data

To prevent data loss:

- Notification data is stored persistently in a database

This database is called:

Notification Log Database

Purpose of Notification Log Database

The notification log database stores:

- Notification events
- Delivery status
- Retry information
- Timestamps
- Event IDs

Benefits

1. Prevents Notification Loss

Even if:

- Worker crashes
- Queue fails
- Service restarts

The notification can still be recovered from the database.

2. Supports Retry Mechanism

If notification delivery fails:

- The system retries later

Because notification data is stored persistently.

3. Failure Recovery

After a system failure:

- Workers can reload pending notifications
- Processing can resume safely

Retry Mechanism

The retry mechanism ensures failed notifications are attempted again.

Examples of temporary failures:

- APNS unavailable
- FCM timeout
- SMS gateway failure
- Network issue

Retry Flow

1. Notification delivery fails
2. Failure status is recorded
3. Notification remains in storage
4. Retry worker attempts delivery again later

Benefits of Retry Mechanism

- Improves reliability
- Handles transient failures
- Reduces notification loss

2. Will Recipients Receive Notifications Exactly Once?

The short answer is:

None

No

Why Exactly-Once Delivery Is Difficult?

Notification systems are distributed systems.

In distributed systems:

- Network failures occur
- Retries happen
- Acknowledgements may be lost
- Servers may crash

Because of these issues:

- Duplicate notifications can occur

Example of Duplicate Notification

1. Worker sends notification successfully
2. Acknowledgement is lost
3. Worker assumes delivery failed
4. Worker retries notification
5. User receives duplicate notification

Goal

The system aims for:

None

At-least-once delivery

Meaning:

- Notifications should not be lost
- Occasional duplicates are acceptable

Reducing Duplicate Notifications

To minimize duplicate notifications:

- A deduplication mechanism is introduced

Dedupe Mechanism

The system tracks notification event IDs.

When a notification event arrives:

Step 1: Check Event ID

The system checks whether the event ID already exists.

Step 2: Decision

If Event Already Exists

- Notification is discarded

Reason:

- It was already processed earlier

If Event Is New

- Notification is processed normally
- Notification is sent to users

Simple Dedupe Logic

None

If event ID exists:

discard notification

Else:

process notification

Benefits of Dedupe Mechanism

- Reduces duplicate delivery
- Improves user experience
- Prevents repeated notifications

Important Note

Even with deduplication:

None

Exactly-once delivery cannot be guaranteed completely.

Due to the distributed nature of the system.

Reliability Summary

Reliability Concern	Solution
Data loss	Notification log database
Temporary failures	Retry mechanism
Duplicate notifications	Deduplication using event IDs
Crash recovery	Persistent storage
Delivery guarantee	At-least-once delivery

Key Takeaways

- Notifications should never be lost
- Persistent storage is mandatory
- Retry mechanisms improve reliability
- Duplicate notifications are possible
- Deduplication reduces repeated deliveries
- Exactly-once delivery is extremely difficult in distributed systems